

Automating the Calculation of Stellar Temperature

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1 Abstract

In 1893, Wilhelm Wien discovered the inverse relationship of the most intense wavelength of light emitted by a star and the star's temperature, which was coined Wien's Law. Though future improvements have been made, Wien's Law remains highly accurate for shorter wavelengths. This project builds on the concept outlined in Wien's Law to automate the calculation of stellar temperature. All data used are synthetic stellar spectra of cooler stars, within a range of 2500°K to 5000°K.

The project centers around finding a viable computer algorithm that accurately computes the most abundant wavelength of stellar spectra, the computer algorithms consisting of maximizing the unaltered, convolved, and regressed data.

Additionally, a new method for curve fitting was proposed, which used Taylor Polynomial Expansions (i.e. Taylor Expansions, TE) of a set of points, whose approximate n th derivatives were calculated. An intermediate product of generating the TE is a list of first derivatives of the stellar data. This data can also be used to find where the data's maximum occurs. Both these methods, however, will need further testing for viability.

All results are compared to approximations calculated by hand. It was found that the three proposed algorithmic methods underperformed when compared to the approximations, where the convolution method performed the best out of the three. However, overall results with the convolved and unaltered data appear scattered and unpredictable, while the human approximations and spline regressions each have specific regions where they under and outperform comparatively.

2 Introduction

The current methods used to calculate the temperature of a star are inefficient. Scientists must measure the star's luminosity, its distance, and its size before beginning their temperature calculations. However, recent innovations in machine learning and computation allow for these calculations to move in a different direction—that of analyzing stellar spectra.

This project is focused on answering two main questions: can the calculation of stellar temperature be automated, and if so which methods can do so reliably.

Temperature is a measure of average kinetic energy, or movement on the atomic level. As a star's temperature increases, so does its kinetic energy. Particulate movement also increases, and does so at higher speed, causing atomic collisions to occur with a greater frequency and force. Electrons' collisions, along with their deexcitation, cause light to be emitted. The type light that is released depends on two main factors: the change in the electron's orbital location and the nucleus of the atom. An electron jumping from its fourth or fifth excited state to its ground state will release far more energy than an electron dropping from its first excited to its ground. The jumps between energy levels are dictated by the Bohr model of an atom, which states that all electrons are confined to specific energy levels, which are said to be quantized, and that there is a lower limit, or the ground state, and upper limit, where the electron thereafter is ionized, or no longer a part of the atom. Each element also emits, or absorbs, their own unique set of wavelengths that are characteristic to themselves, which, when measured, can be used to determine the composition of the star through spectroscopy, or the study of a stellar spectrum.

Stellar spectra measure the abundance of light from a star at each wavelength. There are three main types of spectra: continuous, absorption, and emission. Continuous spectra are arrays of all wavelengths of light that commonly serve as backdrops for absorption spectra, which are similar to continuous spectra yet have dark lines, or dips, caused by cooler atoms absorbing photons and preventing some of them from being detected or measured. The idea that cool gases absorb light is defined by Kirchhoff's Third Law. Absorption happens in stars where their core, which is undergoing nuclear fusion, is releasing light at all wavelengths while its outer shell, filled with cooler gases, are absorbing the energies emitted. Emission spectra are the opposite of absorption spectra—they appear as bright, colored lines on a dark background caused by a specific element gaining enough energy that its electrons excite and deexcite, releasing photons at their element's characteristic wavelengths. This property is defined by Kirchhoff's Second Law. Absorption and emission spectra are inverses of each other, and the dark lines, or dips, of the former coincide with the bright lines, or peaks, of the latter. There are also two common representations of spectra: in graphical form, where we see peaks and dips, and colored form, where we see bright and dark lines. This project uses the graphical form of the data, as it allows for the calculation of the maximum of a dataset in addition to visualizing other relationships, which an array of colors cannot. When referencing the data, x-values and wavelengths will be used interchangeably as well

as y-values and fluxes.

For the purposes of this project, all stars are assumed to be black bodies—ideal radiators who absorb all incoming radiation without reflecting any, meaning all light observed from the body originated from its core. Stars are near black bodies and are commonly reduced to be considered ideal radiators, so it is not too farfetched to make that simplification. However, this can possibly cause large inaccuracies when the computer is handling stars that do not follow closely to black body radiators.

To calculate the temperature of a star using its spectra, Wien’s Law, which dictates the inverse proportionality of temperature and the most abundant wavelength produced by a star, can be employed after finding the occurrence of that wavelength. Wien’s Law can be derived from Planck’s Law, which is defined as:

$$B(\lambda, T) = \frac{2hc^2}{\lambda^5} * \frac{1}{e^{hc/\lambda kT} - 1}$$

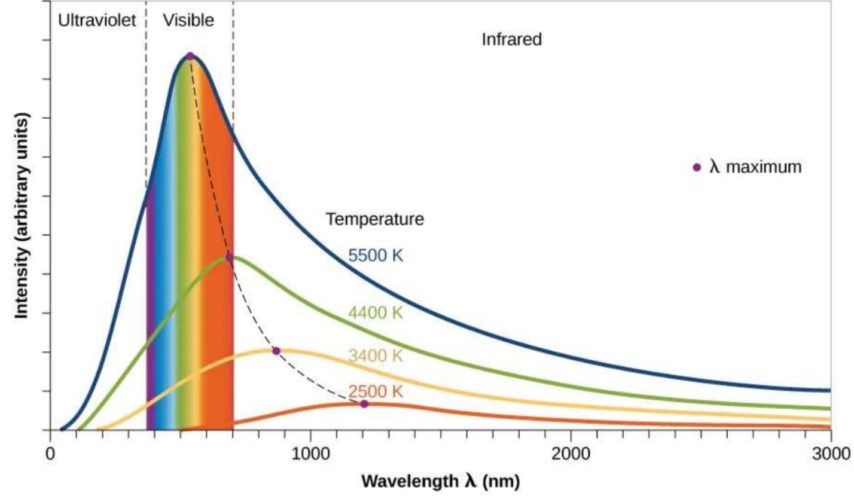
To calculate the temperature of a star using its spectra, Wien’s Law, which dictates the inverse proportionality of temperature and the most abundant wavelength produced by a star, can be employed after finding the occurrence of that wavelength. While Wien’s Law has been expanded upon, the inverse relationship has been upheld by differentiating Planck’s Law and finding where the derivative equals zero. Upon doing so, it has been found that the maximum wavelength occurs at

$$\lambda_{max} = b/T$$

where $b = 2.897771955E7\text{\AA} \cdot K$. This property is only applicable for black bodies—ideal radiators who absorb all incoming radiation without reflecting any, meaning all light observed from the body originated from its core. Stars are near black bodies and are commonly reduced to be considered ideal radiators. Their radiation is approximated by Planck’s Law, which takes both temperature and different wavelengths as its input and outputs the radiance, or amount, of each wavelength the radiator would ideally produce (Figure 1).

This project heavily relies on a star’s stellar spectrum to closely follow the trends outlined by a Planck’s Law curve (PLC). However, that tends to not be the case for hotter stars. While cool stars follow closely to Planck’s Law (Figure 2), hot stars do not. Hotter stars have more movement on the atomic level, meaning they experience more collisions at higher energies. This causes hotter stars produce a larger portion of their light in higher, or bluer, amounts of energy. In addition to that, a star is mostly comprised of hydrogen. Hydrogen atoms absorb light at their own characteristic wavelengths, as dictated by the Lyman series, the Balmer series, etc. Looking at just the Balmer series, it is clear that the majority of light absorbed by hydrogen is in the bluer part of the electromagnetic spectrum, centering around 3000Å to 6000Å. This region coincides with the peak of most hotter stars’ Planck curves, making it difficult to find where the maximum occurs, as there are large drops in the graph (Figure 3). Additionally, these hydrogen atoms, and any other atom for that matter,

Figure 1: shows a set of different Planck curves for different temperature black body radiators. Higher temperature radiators have light that is in the bluer, or more energetic, part of the electromagnetic spectrum while lower temperature radiators experience the opposite, being redder (Fraknoi et al., 2022).



will release the energy it absorbs at some point. However, the light that is released can be refracted to travel in a direction that is not radially outward causing less light to be measured, and can also be reemitted at lower energies over time, causing the graph of the data to spike in its less energetic regions, making it near impossible to accurately detect where the maximum occurs.

3 Data

For the purposes of this research, 84 synthetic stellar spectra were sourced from POLLUX BT-Dusty, an online database of computer generated arrays with specific input parameters (Figure 4). The data's domain spans from approximately 3000Å to 12000Å, containing the necessary information to calculate any spectrum's temperature within the range of 2500 to 10000°K. All temperatures of spectra that were chosen were further limited to be within the range 2500°K to 5000°K. This was done to avoid the issues explored above, where electrons in a star's outer atmosphere absorb, refract, and alter the light that is passed through them by hotter stars.

The actual y-values, or fluxes, of the data were not crucial—just that the overall trends where the maximums occur are still in tact. For that reason, units were not closely examined and axes were not altered to specifically match the units of spectral radiance as the input value units still matched. This data contains no redshift, so no calculations must be done before use in that regard.

Figure 2: shows the spectrum for a cooler star (pink) compared to Planck curves for stars with similar temperatures. The cooler star is modeled well by the the Planck curves (Richmond).

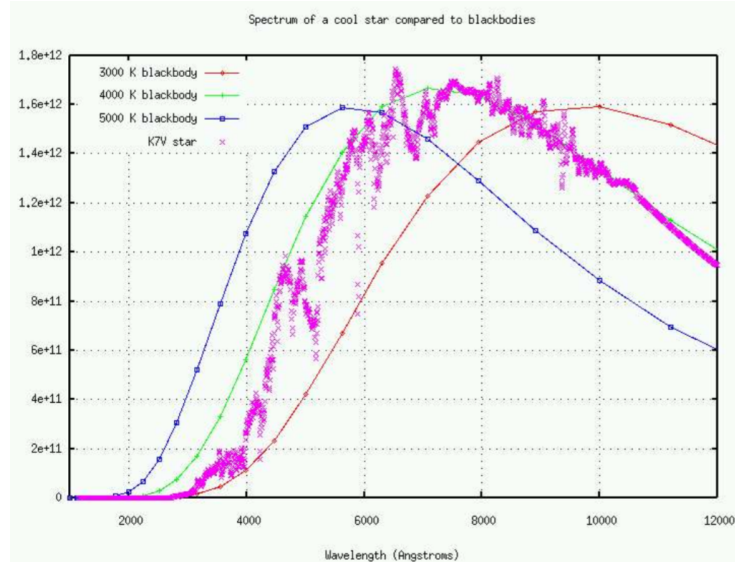


Figure 3: shows the spectrum of a hotter star (red). The data from the hotter star is much more unpredictable and does not follow closely to black body radiators at nearby temperatures (Richmond).

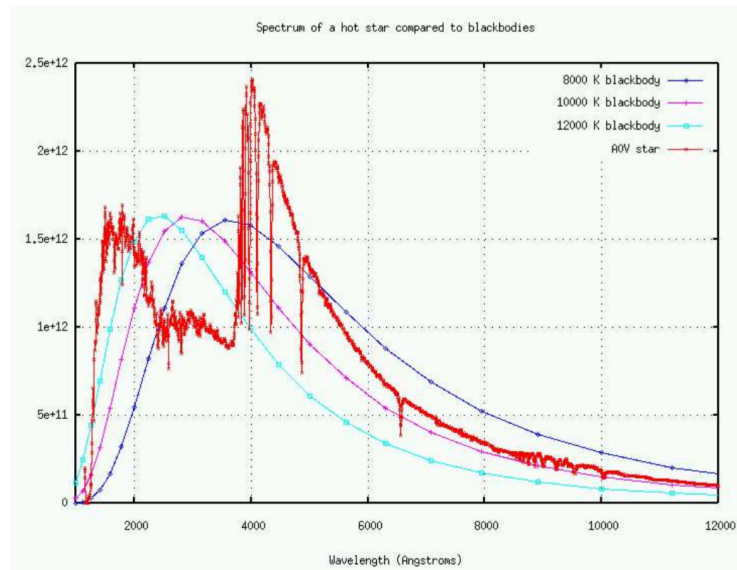
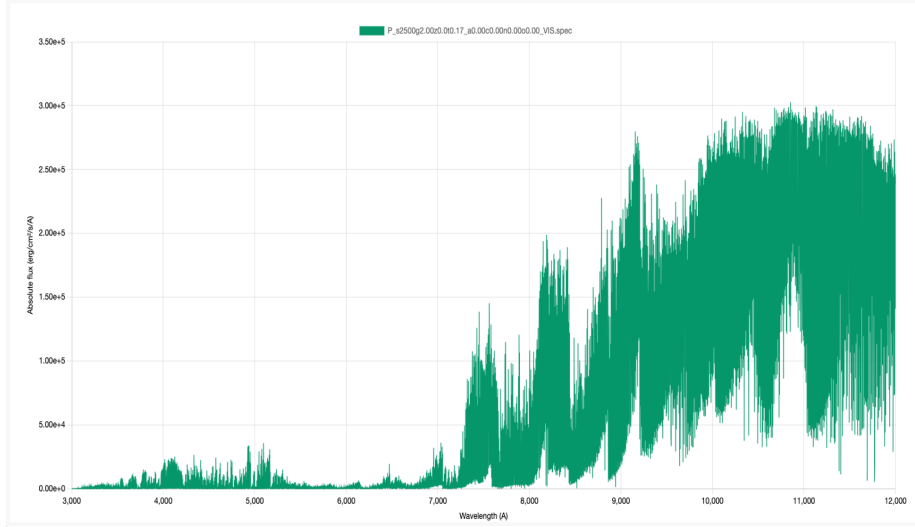


Figure 4: an example of a synthetic stellar spectrum that was sourced from POLLUX.



4 Methods

This project centers on finding the maximums of datasets. To do so, there were three main methods proposed in addition to generating a baseline.

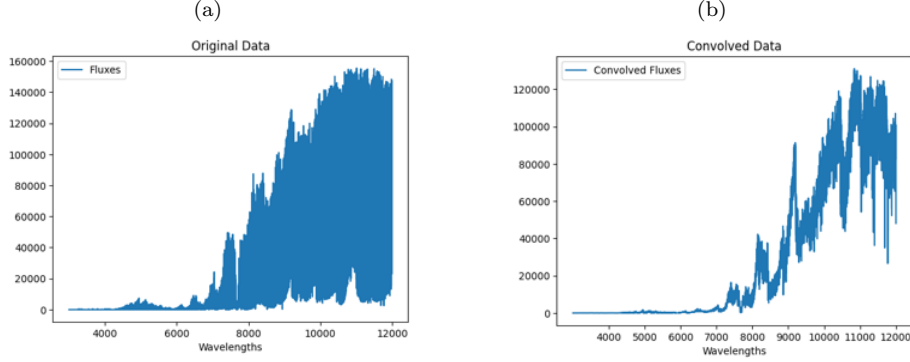
To create the baseline data, the overall trends of the data are visualized, the highest point of the trend is found, and its corresponding x-value, or wavelength, is stored. Then, the wavelength is used in a Wien's Law calculation to find the spectra's temperature. The reasoning behind this is to see if the computer can improve upon calculations done by hand, to make their either more accurate or less time consuming.

Then, before any alterations to the data are made, the computer is programmed to find the largest y-value, or flux, on the spectrum. It finds that maximums corresponding wavelength and using Wien's Law to find the spectrum's temperature.

Additionally, a convolution is taken of the spectrum with an width of 101 indices (Figure 5). This was used to clean the data, making it less noisy and following more to the trends in the data. Examining trends instead of looking at all data points is helpful in this project as Wien's Law, being partly derived from Planck's Law, relies on the spectrum conforming mostly to the shape of the star's respective black body curve. The trends of the data, in theory, more closely follow the overall shape of a Planck curve. Once the convolution is run, the maximum and its corresponding wavelength are found and used to find the spectrum's temperature.

Lastly, a spline regression of the data is taken. A spline regression is a stitching together of multiple polynomial curves at locations called knots. There

Figure 5: the change in the data due to taking a convolution of the original dataset. The convolution (b) has much less noise is more closely resembles a Planck curve than the original data (a).



are two main degrees of freedom in a spline regression: the location/amount of knots (which dictates the number of curves) and the degree of the polynomials. The regression was computed with 7 equally spaced knots, which would mean 8 polynomial, in this case cubic, curves (Figure 6). Once the regression is complete, the maximum was, again, found and its corresponding wavelength was used to find the spectrum's temperature.

Then, the three methods were combined into a single program that allowed for the computer to automatically go through each piece of data to find the temperature predicted by each method. The outputs were saved to a csv file and were analyzed and regressed to look for trends.

4.1 Proposed Methods

In addition to the above methods, a few other methods were proposed. The first was looking at the first derivative of the datapoints and the second consists of finding the Taylor Expansion (TE) of the curve around the maximum and solving the expansion to find the maximum.

However, before any of those methods were tested, another mechanism for data cleaning was tried. This method worked differently from a convolution, where it took averages of intervals and stored them to the center of each interval: interval averaging (IA) method. This helped to simplify data and, again, make it follow more closely to trends in data with less attention to noise (Figure 7).

From there, the first derivatives of the IA data are calculated. Once they are found, it is a matter of finding where the derivative switches from positive to negative (where the original data goes from increasing to decreasing) and setting that to be the location of the maximum (Figure 8). However, in practice that does not work well as the fluctuations, even in the cleaned data, make it nearly impossible to determine where the absolute maximum occurs from just

Figure 6: a spline regression of the dataset used to predict where the maximum of the data occurs.

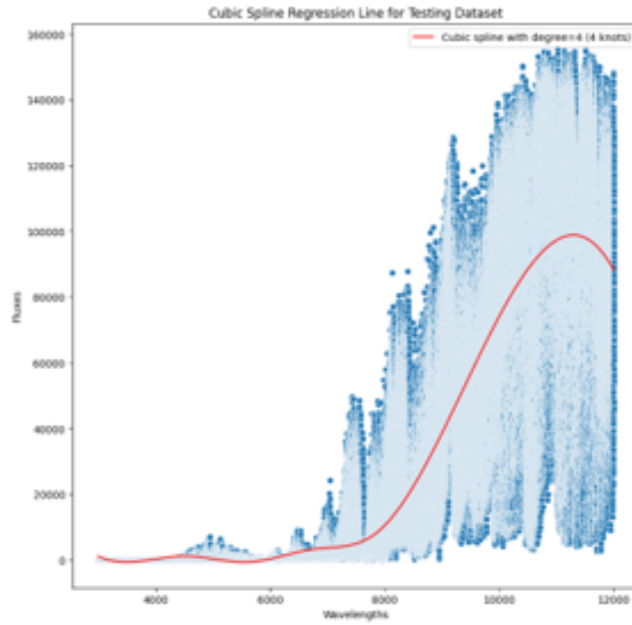


Figure 7: this data is an example of spectral data after the IA method was applied to it, where the data is much simpler and cleaner.

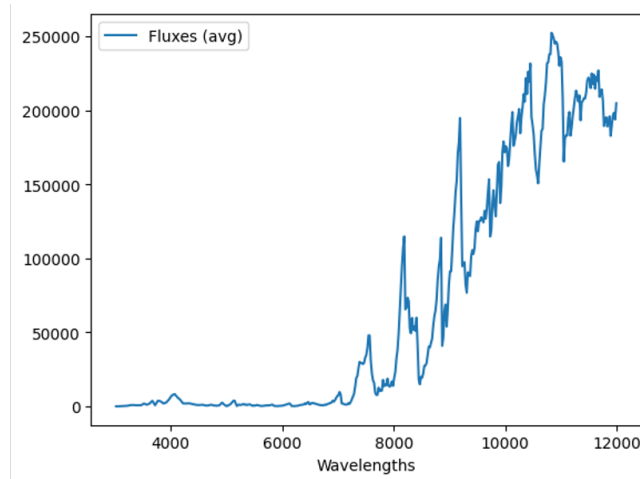
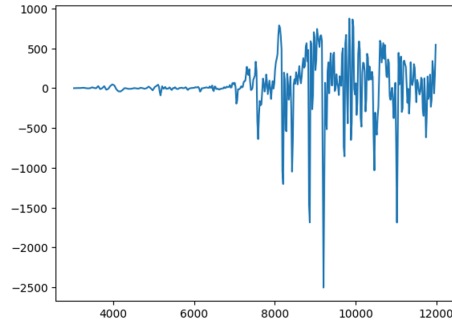


Figure 8: shows an example of the first derivative of the IA data. The derivative is highly volatile and crosses the x-axis frequently, making it difficult to see where the true maximum lies by finding the zeros alone.



that data alone.

The other proposed method is to use a Taylor Series to find a near value approximation of the datapoints close to the location of the maximum. That approximate location can be found through using any of the three methods above, as complete accuracy is not extremely important, just that the general location of the maximum be found. Then, the derivatives of the data up to the seventh degree at that point are calculated. Those values are then plugged into the Taylor Polynomial Expansion formula, which is defined as:

$$T(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x - a)^n$$

On a pseudo dataset used for testing, this method worked extremely well and clearly fit the trends in the data with only 8 terms, including the constant term in the expansion (Figure 9).

The three proposed methods above were used on a piece of real data that the other methods had run on, but the results were poor as the large variability in the data created inconsistencies between the TE, the zeros of the derivative, and the location of the maximum (Figure 10).

5 Results

The three proposed mechanisms did not perform better than the approximation method on average. The baseline data performed with a 7.625% error. In comparison, the method where no alterations were made, the convolution method, and the spline regression performed with a 9.334%, 8.117%, and 9.925% error, respectively (Figure 11). However, when factoring in if the methods over or under approximated the temperature, the convolution method performed very well, scoring a 0.566% error. The calculations done without a computer now scored the worst, with a 6.375% error, consistently over estimating the values of

Figure 9: shows a proof of concept for the TE method. The TE method was tested on a small set of pseudo data and was tasked with fitting a curve around $a = 1$ (black) and $a = 5$ (purple). The closer the value of a is to the maximum, the better job the method does at finding the true maximum of the curve.

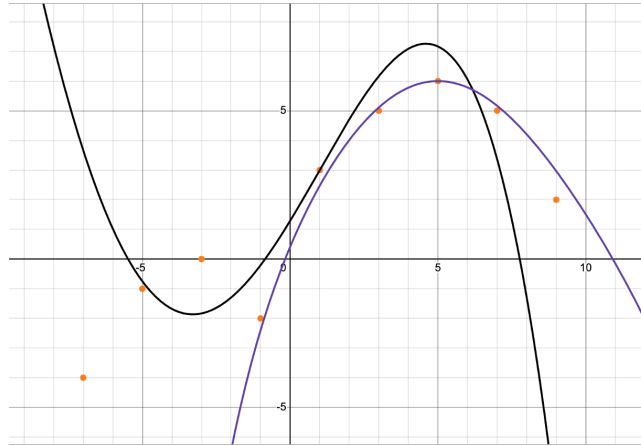


Figure 10: shows the TE for a real dataset, which is grossly inaccurate, having an exponential decay shape in addition to a negative and skewed y-axis.

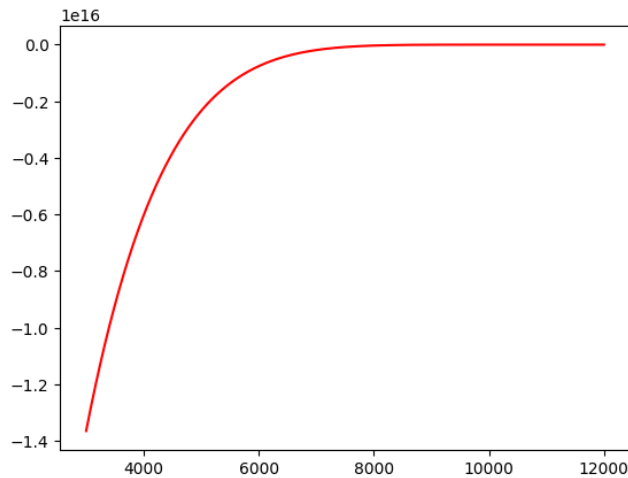


Figure 11: Magnitudes of Percent Errors for each Method at each Temperature



temperature. The unaltered and the spline regression computed with an average of a 3.760% and -5.365% error, respectively (Figure 12).

Then, the data of each were superimposed on each other and regressed with a sixth-degree polynomial, to which the R^2 value was computed (Figures 13 & 14). The data shown is a scatter plot of the magnitudes of percent errors and the true values of percent errors including their signs, respectively. The regressions were performed to look for any general trends or patterns that appear in each method or across methods at specific temperatures.

6 Discussion

The purpose of this project was to automate the processes used to find the temperature of a star. Overall, each methods had an average of less than a 10% error. However, they did not out perform the calculations done by hand, which were still done with extremely approximate values of the most abundant wavelength. Additionally, the baseline was much quicker to generate than the convolution, taking a similar amount of the time to the spline regression and maximum to calculate. This makes sense, as the dataset had over 390,000 indices of data stored, to which the convolution had to go through and add each value 101 times in addition to dividing each sum by the interval width for every index in the original data. The large size of the data was a large inhibitor in scaling the project any further, or using even larger amounts of data with wider domains. In the future, that could be improved upon, where instead of taking

Figure 12: Percent Errors for each Method at each Temperature, including signs to see which methods tend to over/under-approximate and for what temperatures.

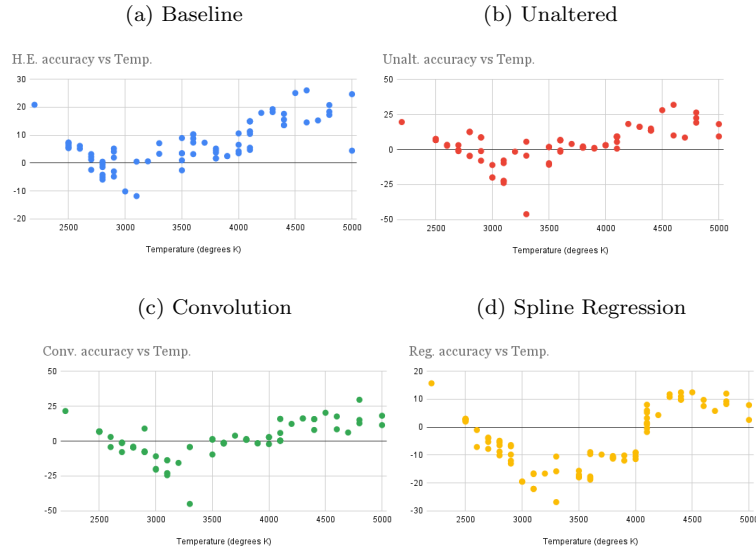


Figure 13: Superimposition of the Magnitudes of all Methods' Accuracies

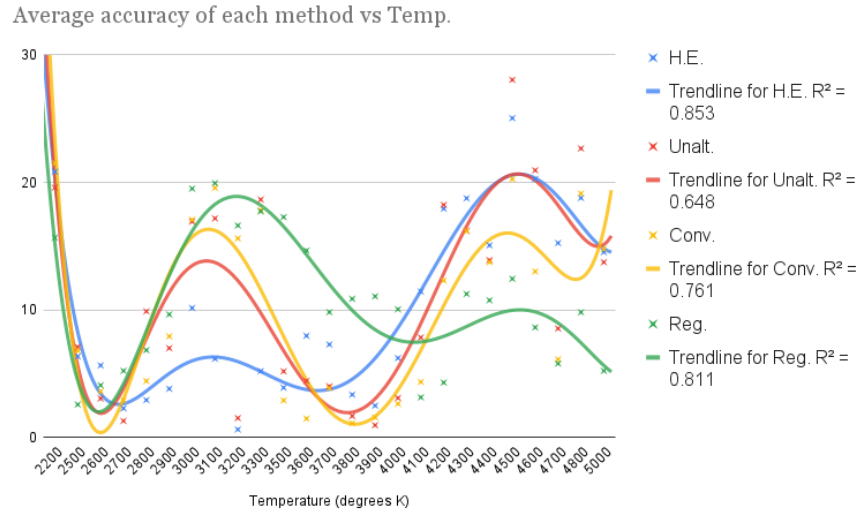
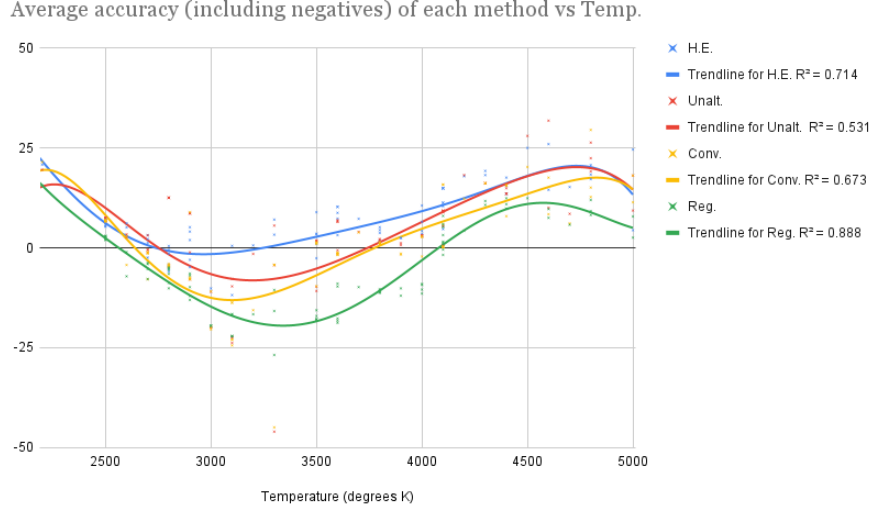


Figure 14: Superimposition of all Methods' Accuracies, with Negative Values



a convolution of all the data, the IA method is applied on a larger scale. This change might also aid the CSR in not overfitting the data, as it removes excess noise.

Overall, yes, the baseline had much greater accuracy in the percent error magnitudes set. But in certain regions, that was not entirely the case. The four methods all trended close together except for in two sections. The first was the baseline method from around 2500°K to 3700°K. Unlike the other methods, the baseline stayed consistently under a 10% error, halving if not quartering the errors of the other methods. This makes intuitive sense. For cooler stars, the data tends to be more uniform and gradual. The human eye, having general context of the curve shapes and trends as well as information about the occurrences and locations of flaws in the data, is a much better tool for temperature calculations when the data is simpler. It can account for anomalies in the data, unlike a rigid maximization or convolution program, and adjust the value of maximum wavelength prediction accordingly.

The second region contains the hotter stellar data, from around 4000°K to 5000°K. In this region, the spline regression, which averaged the worst out of all methods, becomes the most accurate method, predicting with an error consistently below 10%. Comparatively, this is around two-third to over half of the other methods' percent errors, namely the baseline, which performed equally as poor as the unaltered data in this section. The machine learning model performed much better for data that was more likely to be irregular, though still being far out of the range for issues like those depicted in Figure 3 to occur.

The convolution, though having a decent average magnitude of percent error, did not largely change the trend in the graphical data, leading it to perform similarly to the unaltered data. They both did poorly in the regions where the baseline or regression specialized. The convolution might be more valuable as a tool to pair with a further human calculation or a regression of the data instead of only finding the data's maximum.

This discrepancy between averages and performance at different temperatures illuminates the issue with just using an average to represent these outcomes. The spectra that were sourced were heavily skewed towards the cooler end of the range. Out of the 84 spectra that were sourced, around 53, or 63%, of the data was within the range where the baseline performed best, while around 30, or 35%, of the data was within the regression's best-performing range. This skew in the data most likely affected the values of the averages, causing the regression to appear to have performed far worse than the other methods when it in fact did not. This is a limitation with data collected from POLLUX BT-Dusty, and perhaps expanding the set to include more high temperature stars, including those who would experience the hydrogen absorption-reemission problem, would allow for a more representative average value.

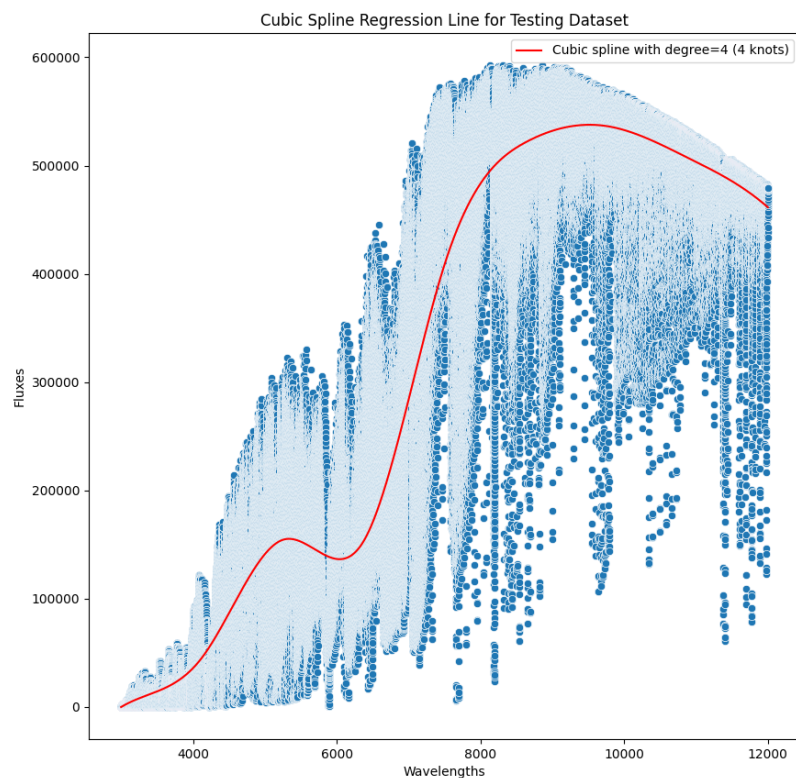
Additionally, as the spectral temperatures increased, the hydrogen atoms' negative effect on the graph is becoming more visible (Figure 15). This causes the data, and more importantly the regression, to have less of a resemblance of a Planck's curve, making it not as feasible to use Wien's Law to find the star's temperature. This issue can be seen in Figure 9, where temperatures nearing around 3000 to 3500°K had the largest percent errors for the regression curve.

Regressions also run the risk of over or under training, where the computer either fits to irrelevant data or misses the fit for important data. With spline regressions, altering the number of knots, their locations, and the degree of the polynomials can affect the shape and accuracy of the regression, and a possible future improvement would be to use an algorithm that primarily optimizes the location of the knots.

In the trend curves for Figures 9 and 10, the lines each have their own R^2 value ranging from 0 to 1, where values closer to 1 show a greater correlation. Both the baseline data and the regression models have higher R^2 values when compared to the convolved and unaltered data, meaning that the average values fit more closely to the trend line. This shows that the accuracy for the approximation and the regression methods are more dependent on the temperature than the original data and convolution. In addition to the lower correlation of the unaltered and convolved methods, the two graphs have a much more sinusoidal trend, unlike the baseline, which can be somewhat approximated with a step function. The fluctuating nature of the convolution and original results begs the question of whether the accuracies are actually dependent on the temperature or if the percent errors are complete random, to which further testing will be needed to draw any conclusions.

When looking at the percent errors that factor in whether the calculation was over or under the accepted effective temperature value, some more interesting conclusions emerge. The spline regression, for the majority of its calculations,

Figure 15: A 3300°K stellar spectrum with a cubic spline regression. This data is exhibiting the issues described earlier, where hydrogen atoms are causing skews early on in the domain of the data, impacting the machine learning model's ability to effectively regress the data. It is clear that the regression does not match the shape of a Planck curve, making it less accurate to apply Wien's Law in this instance.



underestimated the temperature, which coincides with its average percent error of around -5%. Although the convolution’s average error is near zero, it can be explained by the fact that the convolution about equally underestimates the temperature as it does overestimate. Again, the unaltered data is very similar to the convolved, as the convolution did not make any drastic changes to the graph. In these measurements, the baseline seems to have done the worst, but that is mainly because it rarely under approximated the temperature of the spectra, causing the average to be much greater than if it had a more equal split of calculating under and over.

In Figure 10, the R^2 value for the regression is much closer to 1, showing how the temperature of the spectra further affects not only the magnitude of the error but the direction, as well. The unaltered and convolved trend lines have a much lower correlation, showing that they are less dependent on temperature in their performance.

In terms of automation, that aspect of the goal has been achieved. The computer successfully ran through all data when left untouched overnight, and stored all results. The accuracy of these results has been less concretely accomplished, yet there are still promising results, having a percent error within the range of 5 to 10%. In reality, stars glow a certain color, which is a good indication of the general temperature of the star. From there, scientists can use different methods tailored to the situation at hand (e.g. a orange-yellow star will be spectral class K, somewhere in the range of 3500 to 5000°K, which means the regression can be employed as it performs best in that region). Additionally, these methods can serve as initial tests to help generate additional prerequisite data for future methods (i.e. value for a in the TE) as only a reasonable degree of percent accuracy is needed to create a highly accurate TE.

7 Conclusions

The best automated method is the CSR, however, that is only the case for hotter stars, as the human approach still outperformed the computer on cooler data. The spline seems to have the most promise, knowing that it’s accuracy is more dependent on the spectrum’s temperature unlike the convolution and original methods, who had more unpredictable results. In terms of the proposed methods, their results are inconclusive and need further development.

8 Future Work

There are still many improvements that can be made and many possibilities for further research. Adaptations to the input data cleaning methods (i.e. further noise reduction and normalization of spectra) for each method will be made, which should allow for faster and more accurate processing of data.

Additionally, a new algorithm for finding the temperature will be tested, which is based on methods of backpropagation learning used by neural networks.

First the computer will take the CSR of the data. Then, the algorithm will calculate the PLC of a specific temperature and scale it so its peak height coincides with the CSR's. Then the root mean squared error (RMSE) or another relevant error metric will be calculated between the PLC and CSR for specific regions of the data. It might also be useful to compare the nth derivatives of both curves by means of RMSE as well, to see when they have similar slopes, concavities, etc. All the error values are stored in a vector, where lower error values should denote a more accurate fit. This vector is the input into the model. Then, the dot product of an initially arbitrary weight vector and the error vector is taken (note that the weight vector will meet the dimensional requirements so the dot product is possible). The dot product is the overall accuracy score, where a lower score means a more accurate temperature prediction for the PLC. This method is repeated for many different temperature values, which all return an overall accuracy score. Whichever temperature has the lowest score is the temperature that the computer predicts for that star.

For the computer to adjust the weights, the accuracy scores plotted as a function of temperature are compared to an ideal error curve, which is in an upward parabolic shape centered around the actual temperature of the dataset. The actual comparison methods are currently being researched to find how to properly adjust the weights. After weights are adjusted, the model will be generalized to a much larger set of data, and will then be tested on other spectral data which it has not seen before.

It is extremely important that these methods all be tested on true stellar data and not only synthetic spectra. Though the POLLUX database provides phenomenal resources with large amounts of data, databases, like the Mikulski Archive for Space Telescopes (MAST), has data with smaller domains. For that reason, some of the above algorithms might become less viable with this data, which is why testing on data from MAST or of similar origin is imperative.

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